to revolutionise the clean energy industry. They may enable electric cars to run on a battery that's a fifth of the cost and a fifth of the weight of batteries at present available in the market, allowing a car to travel from Delhi to Allahabad, just about 650 km, on a single charge. Today an electric car can run about 300km per charge.

Energy density, that is, the energy that can be extracted from various energy storage materials, is shown in Table 1.

Due to engine inefficiency and heat loss, gasoline is predicted to have a practical lower energy density of 1300Wh $kg^{-1}$ (wastes 70-88% energy). Lithium metal is very light. Li-air battery has a theoretical energy density of 12,000Wh $kg^{-1}$, excluding the oxygen mass. Accounting for the weight of the battery pack, theoretical energy density may reach 3500Wh $kg^{-1}$ for lithium-air battery.

Although energy density is very high, there are practical challenges impeding the development of lithium-air batteries. Previous attempts to develop a practical battery have resulted in low efficiencies and poor performance. However, continuous research work has now yielded a good lithium-air battery in the laboratory, which is about 90% efficient and can be recharged about 1000 times, reaching the end of the tunnel for commercial production.

## Technical challenges

There are many challenges to commercially produce lithium-air batteries. Some of these are described below.

1. A major challenge is the presence of moisture and carbon-dioxide ($CO_2$) in the air, which significantly reduce the cell performance due to their strong affinity with lithium metal. The reaction of lithium with oxygen yields solid lithium-peroxide ($Li_2O_2$) on the cathode (positive electrode) of the battery. Due to low solubility of $Li_2O_2$ in organic electrolyte, it quickly precipitates out of the solution as crystals. These are sometimes deposited in great quantities at the cathode, resulting in sudden death of the cell.

2. Lithium metal on the anode forms dendrites (microscopic fibres of lithium metal), which short-circuit electrical path, and sometimes get electroplated and react with the electrolyte, etc.

3. Lithium metal, besides oxygen in the air, reacts with other components of air like nitrogen, $CO_2$, and water vapour. To get a practical system, it becomes imperative to use filtration to get rid of the other components. The nonaqueous system requires air-purification to reduce water and $CO_2$ content. But no acceptable purification system for EVs has so far been developed. However, aqueous lithium-air batteries can operate in the atmosphere because the discharge products are highly soluble in the catholyte.

4. The lithium anode and the aqueous electrolyte of aqueous lithium-air batteries should be separated by a water-stable lithium-conducting solid electrolyte with high mechanical strength, because lithium metal reacts violently with water. The NASICON (lithium superionic conductor) type solid electrolytes are extensively used. But these become unstable on coming in contact with lithium metal. So, a lithium-conducting electrolyte is used as a buffer layer between lithium anode and the solid electrolyte. The disadvantage of the aqueous system is that the weight and volume of the solid electrolyte separator can reduce the specific energy density of the cell significantly.

**TABLE 1**
**ENERGY FROM SOME ENERGY STORAGE MATERIALS**

| | Theoretical Energy Density (Wh $kg^{-1}$) | Practical Energy Density (Wh $kg^{-1}$) |
|---|---|---|
| Gasoline | 12,889 | 1300 |
| Lead-acid battery | 170 | 20-40 |
| Ni-Cd battery | - | 45-80 |
| Ni-metal hydride battery | - | 60-120 |
| Li-ion battery | - | 100-265 |
| Li-air battery | 3500 | 1300 |

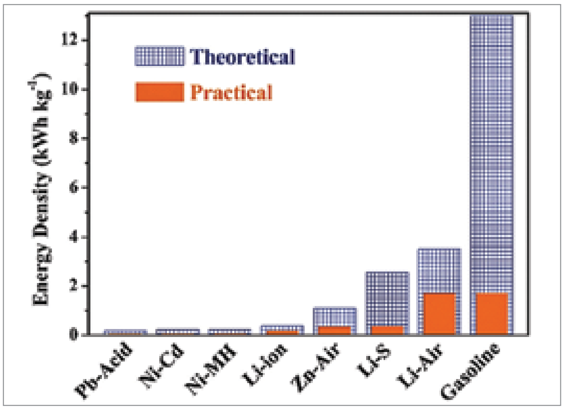


*Fig. 1: Specific energy densities of various rechargeable batteries*

5. There is difficulty in getting the reaction that forms the solid product lithium peroxide to work reversibly.

6. The efficiency is low due to the fact that the difference between discharge and charge voltage is very large, causing lot of energy loss during charging and discharging process.

7. Lithium-air battery using solid-state electrolyte is attractive with respect to safety issues and long-term stability. But stable and dendrite formation-free lithium-ion conducting solid electrolytes are yet to be developed. Hence, a lithium stabilising interlayer, such as a liquid or a polymer electrolyte, is used in lithium-air batteries having solid electrolyte. Further, the cell capacity is dependent on the structure of the air electrode.

8. Power cannot be drawn very fast out of the battery.

## Solution

Some solutions to the problems mentioned above are:

1. Solid lithium-peroxide is a